The Impact of Agency Policies on the Level of Force Used by Officers

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ABSTRACT

Recent high-profile incidents of deadly police use of force (UOF) against civilians have prompted many law enforcement agencies (LEAs) to revise their UOF policies. Evidence suggests that agency policies affect the incidence of agency aggregated counts and rates of UOF, but there is insufficient data showing whether policy influences officer decision-making beyond situational factors. In partnership with the [institution blinded for review], [institution blinded for review] collected data on over 4,000 police UOF incidents involving firearms and less-lethal weapons occurring between 2015 and 2019 from 26 LEAs across the U.S. The present study examined the impact of UOF policies on officer choice of weapon type in UOF incidents among 22 LEAs. Specific policy components examined included accreditation, whether the UOF policy requires attempts at de-escalation, and whether deadly force is permitted for misdemeanors. Officers in accredited LEAs were more likely to use electronic control weapons or conducted energy devices (ECW/CED) than firearms. Officers in LEAs that require consideration of the subject's mental health state were more likely to use chemical agents, CEW/CED, canine, and impact weapons than firearms. The effects of agency policy remain significant after controlling for situational factors. Findings will help LEAs identify the UOF policy components that most impact officer behavior during citizen encounters.

Keywords: use of force, policing, policy, officer weapon type, incident-level data

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INTRODUCTION

An officer's decision to use force, and what type of force to use, during an encounter is impacted by many different factors – individual characteristics of the officer and subject, situational characteristics, agency characteristics, and broader community-level factors. Research shows that situational factors are strong predictors of officer decision-making (Hine, Porter, Westera, Alpert, et al., 2018; Hine, Porter, Westera and Alpert, 2018; Klinger et al., 2015; McElvain and Kposowa, 2008). Agency policy has also been found to be important in guiding officer behavior at the aggregate level (Alpert et al., 2012; Alpert and MacDonald, 2001). It is, however, not well understood to what extent agency policies impact officer decision-making regarding the level of force to use during an encounter above and beyond the influence of officer, subject, and situational factors.

Research that examines the association between agency policies and aggregate counts of use of force (UOF) is limited by the inability to parse out agency policy effects or situational effects without controlling for situational factors. For example, when an agency has a strict UOF policy and low UOF rate, it may partly be because officers in the agency are more likely to be in lower risk situations that require less force. Given officers are the ultimate decision makers during an encounter, it is important to examine the impact of agency policies at the incident level to control for situational factors. Incident-level analysis allows for a more nuanced interpretation of the role that agency policy plays in shaping officer behavior. Agency UOF policies may allow greater discretion by the officer regarding the level of force that can be used and be considered justifiable, which may, in turn, impact officer decision-making during an incident. This poses important implications for public safety, with certain force types being more likely to cause injury or death to subjects than others (Alpert et al., 2011; MacDonald et al., 2009; Smith et al.,

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2007; Stroshine and Brandl, 2020; Taylor and Woods, 2010). The present study examines the impact of agency UOF policies on the type of weapon an officer used from an incident-level database of UOF and officer-involved shooting (OIS) incidents from 22 law enforcement agencies (LEAs) across the U.S. from 2015 to 2019.

Theoretical Framework

Many factors that occur during an incident influence officer decision-making. For example, officers are more likely to use force when the subject uses force (Garner, 1996; Garner et al., 1995) or is armed (Johnson, 2011). While incident-level factors require officers to adapt and react during an evolving situation, agency-level factors, including UOF policy, may influence officer behavior as well. Literature suggests that agencies with more restrictive deadly force policies are associated with less UOF (Alpert, 1989; Alpert and MacDonald, 2001; Fyfe, 1979; Jennings and Rubado, 2017; Sherman, 1983; Terrill and Paoline, 2017). Organizational culture of an agency plays a role in shaping individual officer behavior (Alpert et al., 2012). For example, whether leadership embodies a more "warrior" or "guardian" mindset is passed down to officers and, as a result, influences behavior during civilian encounters. Authority maintenance theory describes that encounters between individuals are governed by a set of exchange rituals that create order (Alpert et al., 2004). Agency UOF policy, in part, determines the set of rituals of the officer and serves as a reflection of agency culture and expectations. Thus, we expect that UOF policies would influence officer decision-making during an encounter. Although there is empirical evidence that agency policy affects the aggregated counts of UOF (Alpert, 1989; Alpert and MacDonald, 2001; Fyfe, 1979; Jennings and Rubado, 2017; Sherman, 1983; Terrill and Paoline, 2017), we do not know whether agency policy influences officer decision making above and beyond situational factors at the incident level.

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The Impact of Agency Policies on Officer Decision Making

Research to date regarding the association between agency UOF policy and UOF has focused on *aggregate* counts of UOF and has found generally that more restrictive UOF policies are associated with less UOF (McLean et al., 2022). Fyfe conducted a review of OIS by the New York City Police Department before and after the introduction of more restrictive shooting guidelines and more intensive incident review procedures and found that there were substantial decreases in fleeing felon shootings, warning shots, and subject injuries after the new guidelines went into effect (Fyfe, 1979). In another study, Shjarback and colleagues conducted a qualitative analysis of 16 years (2003-2018) of OIS incident reports from the Dallas Police Department to assess changes in OIS before the implementation of a more restrictive policy that went into effect in 2013 that requires officers to document brandishing a firearm at citizens (Shjarback et al., 2021). Shjarback et al. found that the policy was associated with a gradual but sustained reduction in OIS.

Agencies may adopt internal oversight (i.e., additional supervision) and external oversight (i.e., accreditation or civilian review boards) mechanisms. Policies regarding internal oversight have found mixed results depending on the policy implemented. Alpert and McDonald analyzed data from a national probability sample of LEAs and found that agencies that require supervisors or other personnel to complete a UOF form rather than the involved officer were associated with lower rates of UOF (Alpert and MacDonald, 2001). Further, findings are inconsistent regarding the impact of body-worn cameras (BWCs) on officer UOF (Lum et al., 2020).

Findings are also mixed regarding the effects of civilian oversight mechanisms in reducing excessive UOF (McLean et al., 2022). A 2018 Department of Justice Community

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Oriented Policing Services (COPS) Office report by Stephens et al. discusses the lack of empirical evidence that demonstrates the effectiveness of civilian oversight programs, and notes the variability in implementation of these programs (Stephens et al., 2018). Little research has studied the impact of accreditation on UOF rates but has not found significant effects to date. Alpert & Macdonald found that CALEA accreditation was not significantly associated with UOF rate (Alpert and MacDonald, 2001), despite the potential benefits of CALEA accreditation to increase agency transparency and community engagement.

The relationship between the UOF continuum and officer UOF is not well understood, but available evidence suggests that more restrictive UOF continua may be associated with less UOF (Bechtel et al., 2023). Most research to date on UOF continua and officer weapon type has focused on the impact of TASERs/conducted energy devices (CED) placement on UOF continua and the impact on TASER use. Findings indicate that more restrictive UOF continua with regard to TASERs are associated with less TASER use (Ferdik et al., 2014; Thomas et al., 2010). When TASERS are placed on a lower tier of the continuum, officers are empowered to use a TASER in place of a less-lethal weapon for a lower level of resistance or less escalation during an incident. Comparing CED use with firearm use, Ferdik et al. found that less restrictive CED policies were associated with fewer fatal shootings (Ferdik et al., 2014). When officers are allowed to use a CED on a lower tier of the continuum, that may reduce the need to further escalate the incident and, in turn, decrease firearm usage. The same logic can be applied to the number of weapons authorized, where it would be expected that a greater number of authorized weapons would be associated with less firearm use. While the UOF continuum is widely adopted by LEAs in the U.S., there is no national standard for a UOF continuum which leads to significant variation

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across agencies regarding the circumstances in which different types of weapons are authorized for use.

There is a large research gap in examining the impact of UOF policies on officer decision-making at the incident level. We identified only one study to date that has examined the impact of UOF policies on officer decision making regarding force used at the incident level, which was limited to three LEAs. Terrill and Paoline conducted an incident-level analysis of 3,340 UOF incidents over a two-year time frame across three agencies to assess the impact of less lethal force policies on UOF, finding that officers from agencies with more permissive policies (i.e., higher levels of force allowed for lower levels of resistance, no linear based UOF model) used force more readily than those from the agency with more restrictive policies (Terrill and Paoline, 2017). Terrill and Paoline controlled for situational variables including subject resistance, subject demographics (race, gender, age), and whether the subject was under the influence of drugs or alcohol, had possession of a weapon, and was deemed to have mental impairment. To note, the authors purposefully selected three agencies with substantive policy variations among eight participating in the study for this analysis. However, the policy variables were not included individually in the model; rather, the authors simply compare the agencies against one another. This analytical approach limits the ability to parse out the impact of individual policies on UOF.

Other Factors that Influence Officer Decision-Making

Officer force type deployed during a UOF incident is influenced by officer, subject, and situational characteristics that interplay and influence the officer's perception of threat, and subsequent response, during the incident. Some research has found that male officers are more likely to use higher levels of force than female officers (Bazley et al., 2007), however, findings

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are mixed (Lawton, 2007; Terrill, 2005). Findings are also inconsistent regarding the association between race and ethnicity of the officer and level of force employed (Lawton, 2007; McElvain and Kposowa, 2008; Sun and Payne, 2004). Officer rank and age are associated with using less lethal force (McElvain and Kposowa, 2008), however, findings regarding officer years of experience are mixed (Alpert et al., 2004; Donner et al., 2017; Paoline and Terrill, 2007; Wheeler et al., 2017; Wright and Headley, 2020). Regarding officer assignment, limited research has also found mixed results (Klinger et al., 2015; Worrall et al., 2018).

Similar to officer characteristics, research indicates that higher levels of force are used on male subjects compared to females and men are more likely to be killed by police (Edwards et al., 2019; Hine, Porter, Westera and Alpert, 2018; Terrill, 2005; Wright and Headley, 2020). Research has found that Black subjects are significantly more likely than White subjects to be shot and killed by police (Jacobs and O'Brien, 1998; Scott et al., 2017; Smith, 2004; Terrill and Mastrofski, 2002; Terrill and Reisig, 2003), while some research has not found significant differences in level of force used and subject race and ethnicity (Garner et al., 2002; Klinger et al., 2015; Shane et al., 2017; Terrill, 2005). Findings are mixed regarding subject age and the level of force used during a UOF incident (Garner et al., 2002; Hine, Porter, Westera and Alpert, 2018; Terrill, 2005). Subjects experiencing a mental health crisis are more likely to have more severe focused used against them compared to those not in crisis (Lawton, 2007), although Johnson (2011) found this association to be non-significant controlling for subject resistance (Johnson, 2011).

Subject behavior during an incident is a strong correlate of officer level of force used. Subject resistance and use of force against the officer are associated with higher levels of force used by the officer (Hine, Porter, Westera and Alpert, 2018; Terrill, 2005; Worrall et al., 2018;

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Wright and Headley, 2020). Additionally, subjects possessing a weapon is associated with higher levels of force (Hine, Porter, Westera and Alpert, 2018; Klinger et al., 2015; Wheeler et al., 2017; White, 2002, 2006; Worrall et al., 2018). Offense type is associated with level of force used, however, there is not a clear positive correlation with force escalating as crimes become more severe. Higher levels of force are used in incidents involving more serious crimes (Lawton, 2007), but calls related to property crimes were also associated with higher levels of force compares to other service calls (MacDonald et al., 2003). Findings are inconsistent regarding the impact of BWCs on force levels (Lum 2020).

The Present Study

In sum, while studies have found UOF policies impact aggregated counts of UOF, we identified only one published study that conducted an incident-level analysis of the association between agency UOF policies and officer force type among, but only among three agencies, and did not analyze the impact of specific UOF policies but rather compared agency-level differences (Terrill and Paoline, 2017). To address the gaps noted above, the current study assessed the relationship between agency characteristics and UOF policies on officer weapon type, controlling for incident-level characteristics of the officer, subject, and situation. Incident-level data of UOF and OIS incidents from 2015-2019 was collected from 26 LEAs across the U.S. varying in agency size, region, and type. Our primary research question was to examine the impact of different agency UOF policies (e.g., if a warning before UOF is required, if a de-escalation attempt is required, if deadly force is permitted for misdemeanor low-level offenses) on the type of weapon used during an encounter. We hypothesized that the likelihood of using more lethal weapons during a UOF incident will differ by agency UOF policies in place. Based on prior research, we hypothesized having a UOF continuum in place would be significantly

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associated with officer weapon type (Ferdik et al., 2014; Thomas et al., 2010). Additionally, we hypothesized that more restrictive UOF policies limiting the circumstances in which officers can use lethal force would be associated with less firearm use compared to other less-lethal weapons (McLean et al., 2022). Findings from the study can be used to inform LEAs when determining what policies may be most impactful on officer behavior during encounters.

METHODS

Sample Description

Incident-level data on UOF and OIS was collected from 26 participating agencies. The sample was drawn from a prior national survey of aggregate counts of UOF, OIS, and violence against officer (OIS) [citation blinded for review] using the National Directory of Law Enforcement Administrators (NDLEA) as the sampling frame. To be eligible for the present study, agencies needed to have at least 100 or more sworn officers and fall within the 80th percentile or above on OIS incidents among the national survey sample. This ensured that participating agencies would have a sufficient sample of UOF and OIS incidents for analysis. A total of 99 LEAs were invited to participate in the study between August 2021 and May 2023, with the final sample of 26 participating agencies. The sample of agencies varied in size, type, and region; four of five regions (using NDLEA region categories) and 14 states are represented. All study activities were approved by the [institution blinded for review] institutional IRB.

Participating agencies provided data for UOF incidents involving a weapon between 2015 to 2019. This included subset of less-lethal UOF incidents and all OIS incidents, given the rarity of OIS. UOF incidents were limited to only those involving the use of a weapon by an officer (e.g., electronic-control device, oleoresin capsicum spray, impact weapon) on a human subject to promote consistency across the agencies. Incidents involving the brandishing a

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weapon alone were excluded. Given variation in what is considered a UOF by agencies across the U.S., using a broad UOF definition would lead to inconsistencies in the data collected and make interpretation challenging. For less-lethal UOF incidents, a random sample was obtained of up to 200 UOF incidents between 2015 to 2019. If the agency had fewer than 200 eligible UOF incidents, they provided all eligible incidents. Data was provided for a total of 4,182 incidents across the 26 agencies. Weights were developed according to the proportion of total cases sampled to represent all UOF cases in these 26 agencies (Pfeffermann, 1993).

For the present study, we merged in data on agency policies from the national survey and from agency UOF policy documents. Policy measures from the national survey were limited, and so the research team requested UOF policy documents from each participating agency to provide richer contextual information. We obtained UOF policy documents for 22 of the 26 participating agencies through the agency providing it directly (12 agencies), the UOF policy being publicly available online (8), or via a Freedom of Information Act request (2). A member of the research team reviewed and coded each agency's UOF policy documents. As such, we had a final sample of 22 agencies for the present analyses. Among the 22 agencies, there were 3,520 incidents: 3,128 including non-lethal weapon UOF incidents and 392 OIS incidents.

Measures

Weapon type

This study examined the type of weapon used by the officer during the officer-subject encounter. Weapon type categories included six initial categories: firearm [reference category], impact weapon, canine, ECW/CED, chemical agent, and other (e.g., patrol car, less lethal munitions)

Agency policies

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CALEA accreditation is the status of accreditation by the Commission on Accreditation for Law Enforcement Agencies, Inc. (CALEA) reported by LEAs and coded as "Yes" or "No" [reference].

Use of force continuum used by the agency was measured as a five-category variable including linear, matrix, circular, or other continuums, or no continuum [reference].

Warning before UOF required was measured as a dichotomous indicator of "Yes" or "No" [reference] if the agency's UOF policy requires that a warning be given before any level of force is used.

Officers required to attempt de-escalation was measured as a dichotomous indicator of "Yes" or "No" [reference] if the agency's UOF policy requires officers to attempt de-escalation.

Deadly force prohibited if the risk is only to self was measured as a dichotomous indicator of "Yes" or "No" [reference] if the agency's UOF policy prohibits deadly force if the subject is only a threat to themselves.

Officers suggested to consider the mental state of the subject was measured as a dichotomous indicator of "Yes" or "No" [reference] if the agency's UOF policy suggests officers consider the mental state, mental health, or cognitive ability of the subject.

Consider pre-seizure conduct was measured as a dichotomous indicator of "Yes" or "No" [reference] if the agency considers the officer's conduct prior to the UOF in assessing the UOF.

Deadly force permitted for misdemeanor low-level offenses was measured as a dichotomous indicator of "Yes" or "No" [reference] if deadly force is permitted against persons thought to have committed only a misdemeanor or other low-level offense.

Use-of-force critical incident review board and *civilian oversight* were measured as a dichotomous indicator of "Yes" or "No" if the agencies have these review mechanisms in place.

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Count of less-lethal weapons authorized is measured as the number of less-lethal tools authorized including baton or other impact weapons, electronic control weapon (e.g., Taser, stun gun), chemical agents (e.g., OC, CS), weapon-deployed chemical agent (e.g., pepper ball), other impact munitions (e.g., rubber bullets, bean bags), neck restraints, and takedown techniques (e.g., straight-arm bar) out of 10 possible options. There was little variation among agencies in the sample (all reporting 7 through 10), and so a three-category variable (7 [reference], 8, or 9-10) was used for analyses.

Body-worn cameras (BWC) deployed was measured as a dichotomous indicator of "Yes" or "No" [reference] if the agency reported that they currently deployed BWCs at the time they responded to the national survey.

Agency-level characteristics

Region was included as a five-category region variable based on the NDLEA categorization, and included Northeast [reference], Midwest, Southeast, South, and West.

Size was measured as a two-category variable for agencies with 100-249 full-time sworn officers [reference] and 250 or more sworn officers.

Type was measured as a two-category variable for municipal [reference] and county agencies.

Violent crime rate in the agency's jurisdiction was measured as the average yearly rate of homicides and aggravated assaults between 2017-2019 per 100,000 population.

Police officer to population ratio reflects the percentage of the number of sworn officers over the population of the jurisdiction the agency serves.

Incident-level characteristics

Incident-level characteristics of the officer included officer *gender*; *age*; *race/ethnicity*; *assignment*; and *rank*. Subject characteristics included subject *gender*; *age*; *race/ethnicity*; if the

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subject *resisted arrest*; if the subject *used force*; *carried a weapon*; was having a *mental health crisis*; and if the subject was *under the influence of drugs or alcohol*. Situational characteristics included reason for the encounter; UCR offense type; if a witness was present; if the incident occurred *indoors or outdoors* (ref.); if *body-worn camera footage* was available for the incident; and if *backup* was requested by the officer.

Analysis Plan

First, we examined the sample descriptives for the incident-level data and the agency-level data. The results are discussed and presented in Tables 1 and 2 in the next section. Two multinomial regression models were then run to compare the effect of each weapon type relative to firearm. For the first model, we fit the multinomial regression model using only incident-level characteristics. We then fit a second model (Tables 3a and 3b), to assess the relationship between agency characteristics and policies on officer weapon type, controlling for incident-level characteristics. The results in the incident-level only model did not change significantly when adding in the agency characteristics and policies, and thus, we do not present the first model results. Prior to the analysis, missing incident-level data were imputed using the *mice* package (van Buuren & Groothuis-Oudshoorn, 2011) in R software, generating 25 completed samples. There were no missing values for the agency-level data. We controlled the clustering by adding agency as a fixed effect in the models. The analyses were conducted using statistical weights to represent all UOF incidents among the 22 agencies (DiSogra, 2009) and were conducted using R software (version 4.2.1).

RESULTS

Descriptive statistics

The dataset comprises 3,520 use-of-force incidents. Table 1 described the incident-level characteristics of these incidents. The incidents involved predominately male (92.2%) and White

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non-Hispanic (69.2%) officers. The average years of service were 9.5 years, and the mean age was 36.1 years. The majority of officers were on patrol duties (78.5%). A large proportion were categorized as Officer, Deputy, or Corporal (87.5%). The use of force incidents predominantly involved electronic control weapons/conducted energy devices (ECW/CEDs) (60.5%), followed by the use of K9 units (12.9%), chemical agents (9.5%), impact weapons (9.1%), firearms (4.6%), and other types of weapons (3.4%). Subjects involved in these incidents were predominantly male (91.1%) and had an average age of 32.4 years. These subjects were most frequently black non-Hispanic (41.3%), followed by white non-Hispanic (34.7%), and Hispanic or other races (24.0%). Notably, a substantial proportion of subjects resisted arrest (77.6%), over half were under the influence of drugs or alcohol (54.8%), and many exhibited signs of experiencing a mental health crisis (35.5%), used physical force (50.2%), or possessed a weapon (33.1%). Unlawful or suspicious activity accounted for the majority of reasons for encounter (54.9%), followed by mass demonstration (14.7%), traffic stops (8.8%), routine patrol (8.0%), service of warrant or court order (7.5%) and medical or welfare assistance (6.1%). Offense type consisted of part 2 crime (54.9%), part 1 violent crime (29.4%), and part 1 property crime (15.7%). Witnesses were present in half of the incidents (50.9%). Most incidents occurred outdoors (63.7%), roughly half of them had body camera footage available (51.1%) and had to request backup (48.3%).

[Table 1 approximately here]

Table 2 displays agency-level characteristics and policies of agencies in the incidents. These incidents were from LEAs across four US regions. The Southeast and West accounted for the largest proportion of agencies at 40.9% each, followed by Midwest (13.6%) and the South (4.5%). About two-thirds (68.2%) of agencies had 250 officers or more. Agency type was

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relatively evenly distributed between municipal and county agencies; slightly over half of the agencies were municipal (59.1%) and 40.9% were county agencies. The average violent crime rate was 261.4 crimes per 100,000 population (SD=222.4). Half of the agencies (40.9%) were CALEA accredited and 86.4% deploy body-worn cameras. Regarding UOF continuum, 18.2% each implement a Linear, Circular, or other model and 22.7% use a Matrix model or no model. Critical incident review boards were present in 18.2% of agencies. Amongst these agencies, almost half (45.5%) used civilian oversight. Close to one in five (18.2%) required a warning before an officer uses force, 81.8% mandated de-escalation attempts (where feasible and appropriate under the circumstances), and 31.8% prohibited the use of deadly force if the risk is only to the individual suspect. About two-thirds (68.2%) of agencies required officers to consider the mental state of the subject in determining the level of force, while 18.2% considered the officer's conduct prior to the UOF in assessing the UOF and 27.3% permitted deadly force for misdemeanors and low-level offenses. Weapon authorization varies: 9.1% of agencies allowed 7 types, 40.9% allowed 8, and 50.0% allowed 9 to 10 types (out of 10 possible). The average number of in-service training hours was 45.3. The distribution of training hours varies: 31.8% of agencies provided less than 25 hours, 18.1% provided between 25 to 39 hours, 31.8% provided 40 to 74 hours, and 18.1%, provided over 75 hours of training.

[Table 2 approximately here]

The association between agency characteristics and policies and officer weapon type

Agency Characteristics

Officers in Midwestern agencies were significantly more likely than officers in Southeastern agencies to use a canine relative to firearms (aOR=6.90). Officers in Southern agencies were more likely to use a chemical agent (aOR=4.77) or an impact weapon (aOR=3.07) than officers

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in Southeastern agencies relative to firearms. Officer weapon type significantly differed by agency size, with officers in agencies with 250 or more full-time sworn officers more likely to use a chemical agent (aOR=3.25) than officers in agencies with 100 to 250 officers relative to firearms. Violent crime rate was associated with increased odds of an officer using a canine (aOR=2.48) relative to firearms. In-service training hours were not statistically significantly associated with officer weapon type.

Agency Policies

Officers in CALEA-accredited agencies were more likely to use an ECW/CED (aOR=1.83) compared to a firearm. The use of force continuum was also significantly associated with officer weapon choice. Officers in agencies with a linear UOF continuum were significantly more likely to use other weapons (aOR=14.52) and an ECW/CED (aOR=3.36) relative to a firearm. Officers in agencies with a circular UOF continuum were more likely to use a canine (aOR=7.26) and an ECW/CED (aOR=2.61) relative to a firearm. Officers in agencies that required a warning before using force were less likely to use ECW/CEDs (aOR=0.44) and canines (aOR=0.09) relative to a firearm. If officers were required to attempt de-escalation, ECW/CED (aOR=2.40) was more likely to be used than a firearm. Officers in agencies with a UOF policy that suggests officers consider the mental state of the subject were more likely to use all other non-firearm weapons, including chemical agents (aOR=6.64), ECW/CED (aOR=1.94), canine (aOR=4.47) and impact weapon (aOR=5.06) relative to a firearm. UOF policies regarding considering pre-seizure conduct, permitting deadly force for misdemeanors, having a UOF critical incident review board or civilian oversight, number of weapons authorized, and body-worn camera deployment were not significantly associated with officer weapon type.

[Tables 3a, 3b approximately here]

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DISCUSSION

The use of force by law enforcement has generated substantial debate, particularly in the last decade following controversial fatal police-citizen encounters. While the law permits officers to use force for safety, the determination of when and how much force is appropriate is governed by a complex framework of legal rules and policies. The core legal concept defining when a use of force becomes excessive arises from the Fourth Amendment of the United States Constitution. In a 1989 case, *Graham v. Connor*, the U.S. Supreme Court established that an officer's use of excessive force constituted a violation of the citizen's Fourth Amendment right to be free from unreasonable seizure (*Graham v. Connor*, 1989). Further, *Graham* outlined the core "reasonableness" inquiry that serves as a foundation for analyzing use of force incidents from a constitutional perspective.[1] While the U.S. constitution provides a minimum threshold level of protection for citizens against excessive force, state constitutions and statutes provide additional protection for citizens by constraining officer's use of force in more specific ways. On top of the threshold requirements established by constitutional and statutory law, agency policies further limit the circumstances in which force is permissible, as well as the type and degree of force that may be employed under the circumstances. UOF policies are intended to reflect contemporary best practices in law enforcement as well as community values and concerns related to law enforcement's use of force.

Due to the hyper-local nature of law enforcement, the structure and content of use of force policies vary substantially from agency to agency. There are differences in how agencies define and explain the threshold legal concepts and there is wide variation in the additional constraints agencies place on the use of force over and above the established legal threshold.

[1] *Id.* (the "reasonableness" inquiry is an objective assessment of whether the officer's actions were reasonable in light of the facts and circumstances confronting them.)

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While other research has explored the influence of situational factors and officer/subject characteristics (Hine, Porter, Westera, Alpert, et al., 2018; Hine, Porter, Westera and Alpert, 2018; Klinger et al., 2015; McElvain and Kposowa, 2008), little research has been conducted to date on the impact of agency policies using incident-level data (Terrill and Paoline, 2017). This study analyzes data from 22 U.S. LEAs to explore the relationship between use of force policy components and the most serious weapon used in encounters involving officer-deployed weapons.

CALEA Accreditation

The incident level data suggests that officers in CALEA-accredited agencies were more likely to use an ECW/CED, a less lethal weapon, compared to a firearm. This finding contradicts that of Alpert and Macdonald; however, their study was not at the incident-level so observed differences may be due to methodology. CALEA accredited agencies must, at a minimum, adopt a standard set of policies, including a specific suite of use of force policies (CALEA, n.d.). CALEA accreditation also influences the frequency, scope and content of UOF training. This finding suggests that the scope and content of CALEA policies may have influenced these officers' choice to use a less lethal ECW rather than a firearm under some circumstances. However, it is also possible that there is "self-selection" bias in CALEA accreditation and that agencies with the accreditation employ policies and training, or otherwise reflect an institutional culture that influences officer choice of weapon in this way.

Use of Force Continuum

Many agencies incorporate a UOF continuum within a UOF policy to provide more explicit guidance to officers regarding what kinds of force are permissible under various threat scenarios (Garrett and Stoughton, 2017). A UOF continuum typically presents a graphic

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representation of the force options permitted based on the conduct of the subject (Ciminelli, 2014). The use of these continua in policy and training is not universally accepted within the law enforcement community, yet many agencies continue to employ them (Garrett and Stoughton, 2017). The main criticism is that these policy components focus almost exclusively on subject resistance and thereby fail to address the nuances that may arise in UOF decision-making (McLean et al., 2023). Another important criticism is that they fail to emphasize potential for de-escalation (Police Executive Research Forum, 2012).

In this study, the inclusion of a UOF continuum as a component of a UOF policy was significantly associated with officer weapon choice, consistent with prior research on CED continuum placement (Ferdik et al., 2014; Thomas et al., 2010). Officers in agencies that use a linear UOF continuum were significantly more likely to use "other weapons" and more likely to use an ECW/CED as the most serious weapon employed during the encounter compared to firearms. Officers in agencies with a circular UOF continuum were more likely to use a canine or an ECW/CED as the most serious weapon employed. Thus, these data indicate that officers influenced by a UOF continuum appear to choose a less lethal weapon (over a firearm), more frequently than officers whose agencies do not employe a continuum, consistent with Ferdik et al.'s findings (2014).

Attempting De-escalation

As de-escalation, where and when possible, has become almost universally accepted as best practice, a majority of agencies have incorporated the concept into UOF training and policy. Officers associated with agencies whose policies suggest or require officers attempt de-escalation when and where possible were more likely to choose an ECW/CED over a firearm as

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the most serious weapon used. This finding is supported by the theory that encouraging officers to consider de-escalation results in their use of less lethal options instead of firearms.

Pre-Use of Force Warning Requirement

Conceivably, an officer warning prior to using force is intended to encourage the subject to cease the resistance or threatening conduct thereby obviating the need for force. Officers associated with these agencies were less likely to use an ECW/CED or a canine relative to a firearm. This counterintuitive result appears difficult to explain. However, one hypothesis might be that officers believe that, once they have complied with the policy's warning requirement, they are justified in using a firearm and need not consider relying on a less lethal weapon.

Consideration of Subject Mental State

Since the introduction of crisis intervention training (CIT) in the 1980s, significant advancements have been made in strategies for managing mental health crises. However, these incidents still pose challenges for law enforcement. Studies show that officers are more likely to use higher levels of force against individuals in mental health crises than against those who are not (Johnson 2011; Lawton 2007). There were two policy variables relevant to this issue that were evaluated in this analysis: (1) whether the agency prohibits the use of deadly force against individuals that are only a threat to themselves; and (2) whether the policy requires the officer to consider the mental state of the subject in determine whether and what type of force to use.

A policy prohibiting the use of deadly force against individuals who are merely a threat to themselves, not the officer or a third party did not generate any observable impact on an officer's choice of weapon in this study. The more prevalent policy prescriptive that requires officers to consider the mental state of the subject, was significantly associated with officer weapon choice. Officers from agencies who require consideration of the subject's mental state

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were more likely to choose to use each type of less lethal weapon (ECW/CED's, chemical agents, canines, and impact weapons) relative to using a firearm, suggesting that the policy may be effective in reducing severity of force.

Limitations

Several limitations should be noted in interpretation of this data. First, while the study is one of the largest incident-level databases known to date and covers county and municipal agencies in four out of five regions in the US, the dataset is not a nationally representative sample as it focused only on large agencies of 100+ sworn officers. Second, because the dataset only includes UOF incidents in which a weapon was used, the analysis of policy variables can only address the potential relationship between policy components and the type of weapon used, rather than the choice to use force at all. Nonetheless, the data demonstrates that certain use of force policy components are correlated with an officer's choice of weapon, and therefore the degree of lethality of the use of force. Third, we assessed the impact of agency policies at the time of the study (2020-2023), on UOF incidents from 2015-2019 and thus cannot conclude that the policies impact officer weapon choice in these incidents. Many of these policies have remained stable over time at the agencies, but it is possible some have changed since the time the incidents occurred.

Implications

In some incidents, UOF by the officer is necessary to achieve the law enforcement objective, whether that is securing a suspect in custody, gaining control of a situation, or protecting citizens or themselves from threatened harm. However, the type and intensity of the force used are important considerations when considering whether the officer's conduct was reasonable under

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the circumstances. The present study showed that variations in certain use of force policy components may be associated with officer weapon choice during an encounter.

Here, we discuss possible implications of these findings for agencies and policing practices.

First, officers employed by agencies that include a UOF continuum in their UOF policy tended to make less-lethal weapon choices, suggesting that incorporating a UOF continuum into a UOF policy has the potential to contribute to reductions in OIS frequency, thereby reducing potential harm to subjects. To note, we did not directly compare UOF continuum types against one another, but rather compared incidents associated with agencies with and without UOF continua in their UOF policies. Additional research is needed to provide guidance to agencies on the most effective type of UOF continuum (Di Nota et al., 2021). As noted by Di Nota et al., most models have not been empirically assessed. Di Nota et al. offer some recommendations on visual UOF continuum models, but further research is needed to assess the level of impact these models have on officer decision making.

Second, a policy that encourages officers to consider the mental state of the subject was significantly associated with an increased likelihood that an officer would use all less-lethal weapon choices relative to firearms. This finding merits attention and perhaps further study given the significant proportion of police-citizen encounters involving individuals experiencing mental health issues; estimates suggest that upwards of 20% of calls for service are related to behavioral health crises (Abramson, 2021; Irwin and Pearl, 2020; Livingston, 2016). It is possible that the agencies that incorporate this policy component also provide more, more frequent, or more impactful crisis intervention training. Nonetheless, our findings suggest that even a policy that merely recommends consideration of important circumstances related to an encounter, without imposing strict requirements, can still influence officer conduct.

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Lastly, a policy requiring officers to attempt de-escalation was significantly associated with a greater likelihood of the officer using an ECW or CED rather than a firearm. This may suggest that incorporation of the concept of de-escalation into UOF policy has the potential to influence officer behavior in the context of more serious incidents, where a higher level of force is required. However, it is important to note that, among the policies examined here, the level of emphasis on de-escalation and the definitions of de-escalation varied substantially. While this finding is promising, de-escalation has yet to receive significant empirical study in the policing field (Engel et al., 2020, 2022), thus, more research is needed regarding which specific de-escalation strategies result in officer behavior change.

Conclusion

This study reveals that agency use of force policies are associated with officer decision-making during encounters, particularly their choice of weapons, among a diverse set of large law enforcement agencies in the U.S. Findings highlight the importance of establishing clear guidelines regarding the weapons and techniques that officers are permitted to use in various circumstances. Future research should explore additional dimensions of agency culture that may also influence the use of force and the types of weapons officers choose.

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REFERENCES

Abramson (2021) Building mental health into emergency responses. *American Psychological Association*, 1 July.

Alpert GP (1989) Police use of deadly force: The Miami experience. In: Roger G. Dunham and Geoffrey P. Alpert (ed.) *Critical Issues in Policing*. Alpert, Geoffrey P. 1989. Police use of deadly force: The Miami experience. In (Roger G. Dunham and Geoffrey P. Alpert, eds.), Critical Issues in Policing. Prospect Heights, IL: Waveland Press. Prospect Heights, IL: Waveland Press., pp. 480–496.

Alpert GP and MacDonald JM (2001) Police use of force: An analysis of organizational characteristics. *Justice quarterly* 18(2): 393–409.

Alpert GP, Dunham RG and MacDonald JM (2004) Interactive police-citizen encounters that result in force. *Police Quarterly* 7(4): 475–488.

Alpert GP, Smith MR, Kaminski RJ, et al. (2011) Police use of force, TASERs and other less-lethal weapons. *Research in Brief, National Institute of Justice, grant number 2005–IJ–CX–0056, May 2011, 1, NCJ 232215.* Epub ahead of print 2011.

Alpert GP, Rojek J and Porter LE (2012) Measuring the impact of organizational culture and climate on police officers' decisions and behaviour. *Canberra, Centre of Excellence in Policing and Security*. Epub ahead of print 2012.

Bazley TD, Lersch KM and Mieczkowski T (2007) Officer force versus suspect resistance: A gendered analysis of patrol officers in an urban police department. *Journal of criminal justice* 35(2): 183–192.

Bechtel T, Ostfeld MC and Shaefer HL (2023) Evidence on Measures to Reduce Excessive Use of Force by the Police. Epub ahead of print 2023.

CALEA (n.d.) Law Enforcement - Standards Titles: CALEA Standards for Law Enforcement Agencies.

Ciminelli ML (2014) Legal Implications of Use-of-Force Continuums in Police Training. *Americans for Effective Law Enforcement (AELE)*. Epub ahead of print 2014.

Di Nota PM, Stoyko P, Jenkinson J, et al. (2021) Critical review of visual models for police use of force decision-making. *Vision* 5(1). MDPI: 6.

Donner CM, Maskaly J, Piquero AR, et al. (2017) Quick on the Draw: Assessing the Relationship between Low Self-Control and Officer-Involved Police Shootings. *Police Quarterly* (2): 213.

Edwards F, Lee H and Esposito M (2019) Risk of being killed by police use of force in the United States by age, race-ethnicity, and sex. *Proceedings of the National Academy of Sciences* 116(34): 16793–16798.

For Peer Review

Engel RS, McManus HD and Herold TD (2020) Does de-escalation training work? A systematic review and call for evidence in police use-of-force reform. *Criminology & Public Policy* 19(3): 721–759.

Engel RS, Corsaro N, Isaza GT, et al. (2022) Assessing the impact of de-escalation training on police behavior: Reducing police use of force in the Louisville, KY Metro Police Department. *Criminology & Public Policy* 21(2): 199–233.

Ferdik FV, Kaminski RJ, Cooney MD, et al. (2014) The influence of agency policies on conducted energy device use and police use of lethal force. *Police Quarterly* 17(4): 328–358.

Fyfe JJ (1979) Administrative interventions on police shooting discretion: An empirical examination. *Journal of Criminal Justice* 7(4): 309–323.

Garner J (1996) *Understanding the Use of Force by and against the Police*. Research in Brief, US Department of Justice, Office of Justice Programs, National Institute of Justice.

Garner JH, Schade T, Hepburn J, et al. (1995) Measuring the continuum of force used by and against the police. *Criminal justice review* 20(2): 146–168.

Garner JH, Maxwell CD and Heraux CG (2002) Characteristics associated with the prevalence and severity of force used by the police. *Justice quarterly* 19(4): 705–746.

Garrett B and Stoughton S (2017) A tactical fourth amendment. *Va. L. Rev.* 103. HeinOnline: 211.

Graham v. Connor (1989) 490 U.S. 386.

Hine KA, Porter LE, Westera NJ, Alpert GP, et al. (2018) Exploring Police Use of Force Decision-Making Processes and Impairments Using a Naturalistic Decision-Making Approach. *Criminal Justice and Behavior* 45(11): 1782–1801.

Hine KA, Porter LE, Westera NJ and Alpert G (2018) Too much or too little? Individual and situational predictors of police force relative to suspect resistance. *Policing and Society* 28(5): 587–604.

Irwin A and Pearl B (2020) *The Community Responder Model: How Cities Can Send the Right Responder to Every 911 Call*. The Center for American Progress. Available at: https://www.americanprogress.org/article/community-responder-model/.

Jacobs D and O'Brien RM (1998) The Determinants of Deadly Force: A Structural Analysis of Police Violence. *American Journal of Sociology* 103(4): 837–862.

Jennings JT and Rubado ME (2017) Preventing the use of deadly force: The relationship between police agency policies and rates of officer-involved gun deaths. *Public Administration Review* 77(2): 217–226.

For Peer Review

Johnson RR (2011) Suspect mental disorder and police use of force. *Criminal Justice and Behavior* 38(2): 127–145.

Klinger D, Rosenfeld R, Isom D, et al. (2015) Race, crime, and the micro-ecology of deadly force. *Criminology & Public Policy* 15(1): 193–222.

Lawton BA (2007) Levels of nonlethal force: An examination of individual, situational, and contextual factors. *Journal of research in crime and delinquency* 44(2): 163–184.

Livingston JD (2016) Contact between police and people with mental disorders: A review of rates. *Psychiatric services* 67(8). Am Psychiatric Assoc: 850–857.

Lum CM, Koper CS, Wilson DB, et al. (2020) Body-worn cameras' effects on police officers and citizens' behavior: A systematic review. *Campbell Systematic Reviews* 16(3).

MacDonald JM, Manz PW, Alpert GP, et al. (2003) Police use of force: Examining the relationship between calls for service and the balance of police force and suspect resistance. *Journal of Criminal Justice* 31(2): 119–127.

MacDonald JM, Kaminski RJ and Smith MR (2009) The effect of less-lethal weapons on injuries in police use-of-force events. *American Journal of Public Health* 99(12): 2268–2274.

McElvain JP and Kposowa AJ (2008) Police Officer Characteristics and the Likelihood of Using Deadly Force. *Criminal Justice and Behavior* (Issue 4): 505.

McLean K, Stoughton SW and Alpert GP (2022) Police Uses of Force in the USA: a Wealth of Theories and a Lack of Evidence. *Cambridge Journal of Evidence-Based Policing* 6(3): 87–108.

McLean K, Alikhan A and Alpert GP (2023) Re-examining the Use of Force Continuum: Why Resistance is Not the Only Driver of Use of Force Decisions. *Police Quarterly* 26(1). SAGE Publications Inc: 85–110.

Paoline EA and Terrill W (2007) Police education, experience, and the use of force. *Criminal justice and behavior* 34(2): 179–196.

Pfeffermann D (1993) The role of sampling weights when modeling survey data. *International Statistical Review/Revue Internationale de Statistique*. JSTOR: 317–337.

Police Executive Research Forum (2012) An integrated approach to de-escalation and minimizing use of force. Author Washington, DC. Epub ahead of print 2012.

Scott K, Ma DS, Sadler MS, et al. (2017) A social scientific approach toward understanding racial disparities in police shooting: Data from the Department of Justice (1980–2000). *Journal of Social Issues* 73(4): 701–722.

For Peer Review

Shane JM, Lawton B and Swenson Z (2017) The prevalence of fatal police shootings by U.S. police, 2015–2016: Patterns and answers from a new data set. *Journal of Criminal Justice* 52: 101–111.

Sherman LW (1983) Reducing police gun use: Critical events, administrative policy and organizational change. In: Maurice Punch (ed.) *Control in the Police Organization*. Cambridge, MA: MIT Press.

Shjarback JA, White MD and Bishopp SA (2021) Can police shootings be reduced by requiring officers to document when they point firearms at citizens? *Injury Prevention* 27(6): 508.

Smith BW (2004) Structural and organizational predictors of homicide by police. *Policing: An International Journal of Police Strategies and Management* (4): 539.

Smith MR, Kaminski RJ, Rojek J, et al. (2007) The impact of conducted energy devices and other types of force and resistance on officer and suspect injuries. *Policing: An International Journal of Police Strategies & Management*. Epub ahead of print 2007.

Stephens DW, Scrivner E and Cambareri JF (2018) *Civilian Oversight of Police in Major Cities*. Office of Community Oriented Policing Services.

Stroshine MS and Brandl SG (2020) The use, effectiveness, and hazards associated with police use of force: The unique case of weaponless physical force. *Police Practice and Research* 21(6). Taylor & Francis: 591–608.

Sun IY and Payne BK (2004) Racial differences in resolving conflicts: A comparison between Black and White police officers. *Crime & delinquency* 50(4): 516–541.

Taylor B and Woods DJ (2010) Injuries to Officers and Suspects in Police Use-of-Force Cases: A Quasi-Experimental Evaluation. *Police Quarterly* 13(3): 260–289.

Terrill W (2005) Police use of force: A transactional approach. *Justice Quarterly* 22(1): 107–138.

Terrill W and Mastrofski SD (2002) Situational and officer-based determinants of police coercion. *Justice quarterly* 19(2): 215–248.

Terrill W and Paoline EA (2017) Police use of less lethal force: Does administrative policy matter? *Justice Quarterly* 34(2). Taylor & Francis: 193–216.

Terrill W and Reisig MD (2003) Neighborhood context and police use of force. *Journal of research in crime and delinquency* 40(3): 291–321.

Thomas KJ, Collins PA and Lovrich NP (2010) Conducted Energy Device Use in Municipal Policing: Results of a National Survey on Policy and Effectiveness Assessments. *Police Quarterly* 13(3): 290–315.

For Peer Review

Wheeler AP, Phillips SW, Worrall JL, et al. (2017) What Factors Influence an Officer's Decision to Shoot? The Promise and Limitations of Using Public Data. *Justice Research and Policy* 18(1): 48–76.

White MD (2002) Identifying situational predictors of police shootings using multivariate analysis. *Policing: an international journal of police strategies & management* 25(4): 726–751.

White MD (2006) Hitting the Target (or Not): Comparing Characteristics of Fatal, Injurious, and Noninjurious Police Shootings. *Police Quarterly* 9(3): 303–330.

Worrall JL, Bishopp SA, Zinser SC, et al. (2018) Exploring Bias in Police Shooting Decisions With Real Shoot/Don't Shoot Cases. Crime & Delinquency. *Crime & Delinquency* 64(9): 1171–1192.

Wright JE and Headley AM (2020) Police Use of Force Interactions: Is Race Relevant or Gender Germane? *The American Review of Public Administration* 50(8): 851–864.

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Table 1. Weighted Sample Descriptives (N=3,520 incidents)

Variable	**Frequency/Mean (SE)**
Officer	
Gender	
Male	92.2%
Female	7.8%
Race\Ethnicity	
White Non-Hispanic	69.2%
Black Non-Hispanic	9.6%
Hispanic/other race	21.2%
Years of Service	9.5 (1.4)
Age	36.1 (2.0)
Assignment	
Patrol	78.5%
Other	21.5%
Rank	
Officer/Deputy/Corporal	87.5%
Other	12.5%
Force Type	
Other	3.4%
Chemical Agent	9.5%
ECW/CED	60.5%
K9	12.9%
Impact Weapon	9.1%
Firearm	4.6%
Subject	
Gender	
Male	91.1%
Female	8.9%
Age	32.4 (1.2)
Race\Ethnicity	
White non-Hispanic	34.7%
Black non-Hispanic	41.3%
Hispanic/other race	24.0%
Resisted Arrest	77.6%
Possessed a Weapon	33.1%
Subject Used Physical Force	50.2%
Experienced Mental Health Crisis	35.5%
Under the Influence of Drugs or Alcohol	54.8 %
Situational	
Year	

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Variable	**Frequency/Mean (SE)**
2015	17.4%
2016	20.5%
2017	20.1%
2018	20.0%
2019	22.0%
Reason for Encounter	
Unlawful/Suspicious Activity	54.9%
Routine Patrol	8.0%
Traffic Stop	8.8%
Service of Warrant/Court Order	7.5%
Mass Demonstration	14.7%
Medical/Welfare Assistance	6.1%
Offense Type	
Part 2 Crime	54.9%
Part 1 Property Crime	15.7%
Part 1 Violent Crime	29.4%
Witnesses Present	50.9%
Location	
Indoor	36.3%
Outdoor	63.7%
Body Camera Footage Available	51.1%
Backup Requested	48.3%

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Table 2. Agency-Level Characteristics and policies (n=22 LEAs)

Characteristics	**Frequency/Mean (SD)**
Region	
Midwest	13.6%
Southeast	40.9%
South	4.5%
West	40.9%
Size	
100-249	31.8%
250+	68.2%
Type	
Municipal	59.1%
County	40.9%
Violent crime rate	261.4 (222.4)
Policies	
CALEA accreditation	40.9%
Body-worn camera deployment	86.4%
Use of Force Continuum Used	
Linear	18.2%
Matrix	22.7%
Circular	18.2%
Other	18.2%
None	22.7%
Use-of-force critical incident review board	18.2%
Civilian oversight	45.5%
Warning before UOF required	18.2%
Officers required to attempt de-escalation	81.8%
Deadly force prohibited if the risk is only to self	31.8%
Officers suggested to consider the mental state of the subject	68.2%
Consider pre-seizure conduct	18.2%
Deadly force permitted for misdemeanor low-level offenses	27.3%
Number of weapons authorized	
7	9.1%
8	40.9%
9-10	50.0%
Training	
Number of in-service training hours	45.3 (28.8)
Training hours	
<25	31.8%
25-39	18.1%
40-74	31.8%
75+	18.1%

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Table 3a. Multinomial logistic regression to examine the effect of agency and incident factors on officer UOF decision making: agency factors[a] (N=3,520)

Characteristic	**Other**	**Chemical Agent**	**ECW/CED**	**Canine**	**Impact Weapon**
	aOR	aOR	aOR	aOR	aOR
Agency Characteristics					
Region[b]					
West	0.85	1.68	1.01	0.44	0.67
Midwest	6.58	0.39	1.62	6.90**	0.62
South	0.79	4.77**	3.17	2.07	3.07*
Size[c]					
250+	3.40	3.25**	1.38	2.09	1.86
Type[d]					
County/State	1.15	0.56	0.74	1.78	0.52
Violent crime rate	0.10	0.88	0.99	2.48*	1.03
In-service training hours	0.99	0.99	1.00	0.99	1.01
Policies					
CALEA Accreditation	0.31	0.60	1.83*	1.05	0.56
Use of force continuum used[e]					
Linear	14.52*	2.41	3.36**	3.21	1.89
Matrix	0.17	2.22	1.27	1.16	1.28
Circular	1.54	0.94	2.61*	7.26**	0.25
Other	0.81	0.63	0.75	0.60	0.48
Warning before UOF required	1.47	0.54	0.44*	0.09***	0.65
Officers required to attempt de-escalation	0.29	1.65	2.40*	1.32	4.13
Deadly force prohibited if the risk is only to self	2.86	1.25	0.92	0.78	1.60
Officers suggested to consider the mental state of the subject	1.39	6.64**	1.94*	4.47**	5.06*
Officers consider pre-seizure conduct	1.72	1.45	1.16	3.10	0.73
Deadly force permitted for misdemeanor low-level offenses	0.76	0.87	1.16	3.95	1.67
No use-of-force critical incident review board	0.97	0.64	0.83	0.86	0.44

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Characteristic	**Other**	**Chemical Agent**	**ECW/CED**	**Canine**	**Impact Weapon**
	aOR	aOR	aOR	aOR	aOR
Civilian oversight	2.94	2.20	1.57	0.60	0.88
Number of weapons authorized[f]					
8	3.02	0.80	2.20	2.46	1.25
9-10	0.29	0.87	1.42	1.08	0.59
Body-worn camera deployment	0.67	0.52	1.29	0.86	1.61

Reference categories: [a]Firearm; [b]Southeast; [c]100-249 full-time sworn officers; [d]Municipal; [e]No continuum; [f]7. Significant values: .p <0.10, *p < 0.05; **p < 0.01; ***p < 0.001.

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Table 3b. Multinomial logistic regression to examine the effect of agency and incident factors on officer UOF decision making: incident factors (N=3,520)[a]

Characteristic	**Other**	**Chemical Agent**	**ECW/CED**	**Canine**	**Impact Weapon**
	aOR	aOR	aOR	aOR	aOR
Officer					
Gender[b]					
Female	0.99	1.01	1.57	0.78	1.09
Race\Ethnicity[c]					
Black Non-Hispanic	1.04	1.50	1.08	0.05**	0.88
Hispanic/other race	0.67	1.06	1.00	0.97	0.99
Year of Service	1.00	0.99	0.99	1.03	1.00
Age	1.00	1.00	0.99	1.01	0.99
Assignment[d]					
Other	0.18***	0.40**	0.29***	1.90*	0.44*
Rank[e]					
Other	2.31	1.12	0.82	0.79	1.70
Subject					
Gender[b]					
Female	3.21*	8.77***	3.21**	2.76*	2.16
Age	0.99	0.98	1.00	0.97*	1.00
Race\Ethnicity[c]					
Black Non-Hispanic	0.59	1.74	1.43	0.92	1.04
Hispanic/other race	0.79	1.76	1.19	0.90	0.83
Resisted arrest	1.67	5.48***	4.92***	1.96	5.88***
Subject was Perceived to Carry a Weapon	0.10***	0.01***	0.03***	0.03***	0.03***
Used Physical Force	0.15***	0.19***	0.26***	0.10***	0.32**
Experienced Mental Health Crisis	1.25	0.53	1.32	0.34*	1.34
Under the Influence of Drugs or Alcohol	0.64	0.74	0.86	0.45*	0.87
Situation					

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Characteristic	Other	Chemical Agent	ECW/CED	Canine	Impact Weapon
	aOR	aOR	aOR	aOR	aOR
Reason for Encounter[f]					
Routine Patrol	0.31	1.90	1.05	0.67	1.18
Traffic Stop	0.78	0.44	0.50	0.41	0.62
Service of Warrant/Court Order	0.38	0.41	0.32*	0.36*	0.34
Mass Demonstration	0.55	0.69	0.71	0.20*	1.27
Medical/Welfare Assistance	0.76	0.64	0.47	0.30*	0.80
Offense Type[g]					
Part I Property Crime	0.72	0.24*	0.37*	1.00	0.31*
Part I Violent Crime	0.21***	0.23***	0.18***	0.25***	0.24***
Witnesses Present	0.25**	0.79	0.47*	0.17***	0.38*
Location: Outdoor[h]	1.07	0.54	0.55	0.74	0.64
Body Camera Footage	1.50	0.93	0.95	1.38	0.56
Backup Requested	0.47	0.56	0.39*	1.15	0.45

Reference categories: [a]Firearm; bMale; [c]White Non-Hispanic; [d]Patrol; [e]Officer/Deputy/Corporal; [f]Unlawful/Suspicious Activity; [g]Part II Crime; [h]Indoor. Significant values: .p <0.10, *p < 0.05; **p < 0.01; ***p < 0.001.

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